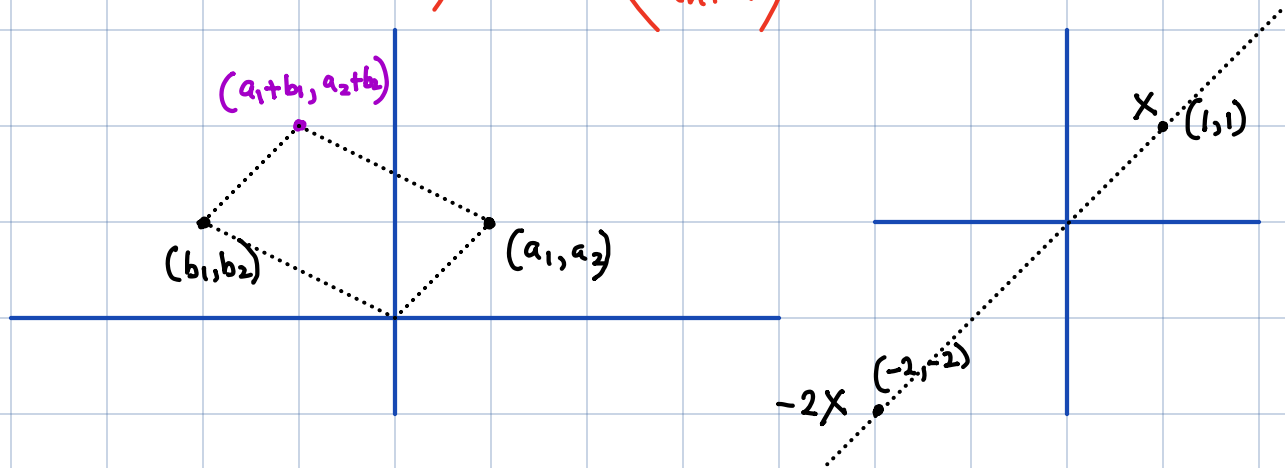


Some Properties of $\mathbb{R}^n$

① One can add or subtract points of $\mathbb{R}^n$

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$$



② For any element $\alpha \in \mathbb{R}$ and a point $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$,

we have $\alpha X = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix} \in \mathbb{R}^n$.

What we shall do now is abstract these ideas and create a "Coordinate-free" Geometry.

Abstract Group

Properties

Let V be a nonempty set having the following

(G₁) Any two elements of V can be "added".

addition $\longrightarrow +$

(G₂) For any three elements $x, y, z \in V$,

Associativity property

$$(x + y) + z = x + (y + z)$$

(G₃) There is an element $0 \in V$ such that

$$x + 0 = x = 0 + x, \text{ for any } x \in V$$

0 is called the identity element

(G₄) For any $x \in V$, there is a $y \in V$

such that $x + y = 0 = y + x$.

In this case, y is called the inverse of x . We denote y by $-x$

Then $(V, +)$ is called a group.

A Group $(V, +)$ that also has the following property

(G₅) for any $x, y \in V$, $x + y = y + x$.

is called an abelian group.

* Caution: $+$ should not be confused with usual addition.
 $+$ is called group operation

Examples

(abelian group) ① $(\mathbb{Z}, +)$ integers with usual addition $+$

(abelian group) ② $(\mathbb{R}, +)$ Real numbers with usual addition $+$

(abelian group) ③ $(\mathbb{R} \setminus \{0\}, \times)$ non zero real numbers with multiplication.

⑥₂ identity element is 1, ⑥₃ for any $x \in \mathbb{R} \setminus \{0\}$, the inverse of x is $1/x$, ⑥₄ clearly, $xy = yx$ for any real numbers $x, y \in \mathbb{R}$.

Definition of a Vector Space: Let $F = \mathbb{R}$ or $\mathbb{Q}$ or $\mathbb{C} \rightarrow$ "Complex numbers"
(Not important)

and $(V, +)$ be an abelian group. If there is a multiplication $*$ between elements of F and V such that
 $f * v \in V$ for any $f \in F$ and $v \in V$

and such that

(V1) $1 * v = v$ for all $v \in V$

(V2) $\alpha * (u + v) = \alpha * u + \alpha * v$, for all $\alpha \in F$, $u, v \in V$

(V3) $(\alpha + \beta) * v = \alpha * v + \beta * v$, for all $\alpha, \beta \in F$ and $v \in V$

(V4) $(\alpha\beta) * v = \alpha(\beta * v)$, for all $\alpha, \beta \in F$ and $v \in V$.

Then V is called an F -Vector Space. (or Vector Space over F).

Note: * (V2) and (V3) are called distributive laws.

* The multiplication $*$ is called scalar multiplication

* (V4) is called associativity of scalar multiplication.

* Often we omit $*$ and simply write αv for $\alpha * v$

Examples

① $\mathbb{R}^n$ is an $\mathbb{R}$ -Vector Space

② Let $(V, +)$ be the set of all 2×2 matrices over real numbers with usual matrix addition.

(i) Prove that $(V, +)$ is an abelian group

(H.W)

(ii) Prove that V is a $\mathbb{R}$ -Vector Space with scalar multiplication

$$\alpha * \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{pmatrix}$$

③ Let $(V, +)$ be the set of all $n \times n$ matrices over real number with usual matrix addition. Prove that V is an $\mathbb{R}$ -Vector Space with scalar multiplication

$$\alpha * (a_{ij}) = (\alpha a_{ij}).$$

Properties of Vector Spaces

① $0 * v = 0$ for all $v \in V$

Solution: $0+0=0$ in F implies $(0+0)*v = 0*v$.

Therefore $(0*v) + (0*v) = (0+0)*v = 0*v$

Let w be the inverse of $0*v$. Then,

$$\begin{aligned} & \underbrace{w + (0*v)} + (0*v) = w + (0*v) \\ \Rightarrow & 0 + (0*v) = 0 \\ \Rightarrow & \boxed{0*v = 0} \end{aligned}$$

② For any $v \in V$, $-1 * v = -v$

Solution

$$1 + -1 = 0 \Rightarrow (1 + -1) * v = 0 * v = 0$$

$$\Rightarrow 1 * v + -1 * v = 0$$

$$\Rightarrow v + -1 * v = 0.$$

Therefore the inverse of v is $-1 * v$ □

Notation for
inverse of
 v .